

Synthesis of a class of new phosphine-oxazoline ligands and their applications in palladium-catalyzed asymmetric addition of arylboronic acids to isatins

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ABSTRACT

A class of new phosphine-oxazoline ligands was prepared from commercially available and inexpensive starting materials. As the first example, in situ formed catalysts generated from chiral phosphine-oxazolines and Pd(OAc)₂ were successfully applied to the asymmetric addition reactions of arylboronic acids to isatins, and 3-aryl-3-hydroxyoxindoles were obtained in up to 97% yields and good enantioselectivities (up to 88% ee). Up to 98% enantiopurity of the products were obtained after one recrystallization.

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1. Introduction

Oxazoline moieties are ubiquitous and privileged structural elements of chiral ligands [1]. Chiral oxazolines are one of the most successful and versatile ligands in asymmetric catalysis [2]. Since the first phosphine-oxazoline ligands (PHOX) were synthesized by Pflatz, Helmchen and Williams in 1993 [3–5], derivatives of these ligands have been continuously developed by introducing different chiral elements into the ligand backbones [2,6]. Many novel chiral phosphino-oxazoline ligands such as PHOX derivatives [2,7–14], spiro backbone phosphino-oxazoline [15,16], chiral axis-fixed biaryl phosphine-oxazoline ligands [17–25], ferrocenylphosphine-oxazoline (Fc-PHOX) [26] and StePHOX [27] were successfully prepared and applied in metal-catalyzed asymmetric reactions.

Chiral 3-substituted 3-hydroxyoxindoles structural cores with a quaternary carbon center at the 3-position are encountered in a large variety of natural products with a wide spectrum of biological

and pharmacological activities [28,29], such as dioxibassinines [30], TMC-95A [31], donaxaridine [32] and convolutamydine A [33]. Studies on the structure–activity relationship showed that configuration of the quaternary carbon stereocenter and substituent of the oxindole has profound influence on the biological and pharmacological activities of those compounds [34]. Catalytic asymmetric addition of nucleophiles to isatins is the most straightforward and powerful approach to this structural scaffold. Some successful asymmetric reactions have been developed in recent years [35–38]. In 2006, Hayashi et al. reported the first catalytic asymmetric addition of arylboronic acids to isatins by using Rh/(R)-MeO-MOP complex as the catalyst [35], and the adducts were produced in high enantioselectivities and excellent yields. Minnaard and co-workers employed a phosphoramidite–rhodium catalyst to this reaction and obtained 3-phenyl-3-hydroxyoxindole in 99% yield and 55% ee [36]. Since rhodium is expensive and rare, replacement of rhodium with cheaper metal such as palladium for this reaction is warmly expected and promising from academic viewpoint and practical application. However, up to now only limited results were reported by employing biphenyl phosphinoimine [39] or N-heterocyclic carbene [40] as the ligand for palladium-catalyzed asymmetric arylation of isatins. Development of more effective ligands and suitable catalytic systems for this

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reaction remains significant and challenging [41]. Axially chiral biaryl phosphine-oxazoline has a structure similar to the biphenyl phosphinoimine and *N*-heterocyclic carbene ligands [39,40]. Moreover, it owns an easily adjustable modular oxazoline moiety. On the basis of the effectiveness for the latter ligands in the reaction, we selected phosphine-oxazoline–palladium catalytic system as our study. Herein, we disclose the synthesis of H₈-BINOL-derived phosphine-oxazoline ligands and their applications in palladium-catalyzed asymmetric addition of arylboronic acids to isatins. The addition reactions acquired good yields and enantioselectivities under mild conditions.

2. Experimental

2.1. General considerations

Anhydrous solvents were dried and distilled according to standard laboratory methods under an atmosphere of nitrogen prior to use. Aryl boronic acids were purchased commercially or prepared according to the literatures and purified by passing through a silica gel column before used. If not mentioned, all other reagents were purchased from commercial sources and used without further purification. All moisture sensitive reactions were carried out under an inert atmosphere of dry nitrogen and were monitored by TLC. ¹H NMR, ¹³C NMR and ³¹P NMR spectra were recorded on a Varian Mercury-Plus 300 spectrometer at 300, 75 and 121.4 MHz, respectively. Chemical shifts (δ) are given in ppm and are referenced to residual solvent peaks (¹H NMR and ¹³C NMR) or to an external standard (85% H₃PO₄, ³¹P NMR). HR-MS was carried out on a Bruker APEX 47e ESI FT-ICR mass spectrometer and Thermo MAT95XP EI-FAB-Cl mass spectrometer. Optical rotations were recorded on a Perkin-Elmer Model 341 polarimeter. HPLC analysis was performed on an Agilent 1200 series system.

2.2. General procedure for the catalytic asymmetric addition of arylboronic acids to isatins

A solution of Pd (OAc)₂ (2.25 mg, 0.01 mmol) and (*S_aS*)-**8g** (9.78 mg, 0.012 mmol) in THF (1.0 mL) was stirred for 30 min at room temperature. KOH (11.2 mg, 0.2 mmol), isatins (0.20 mmol), arylboronic acids (0.4 mmol) and additives (0.03 mmol) were added successively with additional THF (1.0 mL). The resulting mixture was stirred for 48 h at room temperature. Then the solvent was removed under vacuum and the residue was purified by flash column chromatography using silica gel with ethyl acetate/petroleum ether as eluent.

3. Results and discussion

3.1. Synthesis of ligands **8**

The concise synthetic route to the phosphino-oxazoline ligands **8** was shown in Scheme 1. Most of the reactions proceeded in excellent yields, and the final products were obtained readily from (*S*)-(-)-1,1'-bi(2-naphthol) in eight steps. Catalytic hydrogenation of (*S*)-BINOL (Pd/C, EtOH) and subsequent recrystallization from *n*-heptane provided (*S*)-H₈-BINOL in >99% ee [42]. This compound was then treated with triflic anhydride (CH₂Cl₂, pyridine, 0 °C) to form **2** in 91% isolated yield [43]. Bis(triflate) **2** was exposed to carbon monoxide (1 atm) and an excess of methanol in the presence of palladium acetate (15 mol%) and 1,3-bis(diphenylphosphino)propane (dppp; 15 mol%) to give monocarbonylation product **3** in 63% yield [43]. The aryl phosphine oxide moiety was introduced by reacting **3** with Ar₂P(O)H in DMSO at 120 °C in palladium acetate (10 mol%),

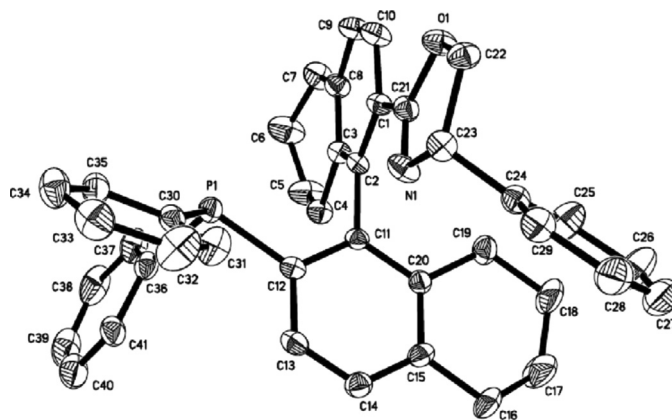
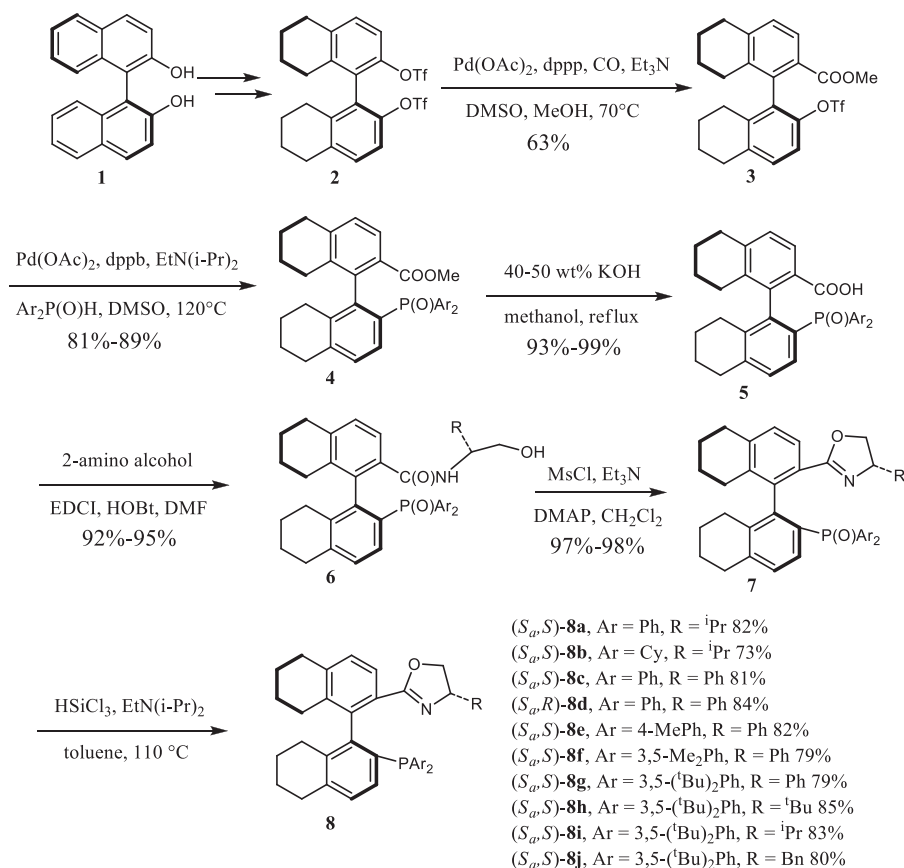


Fig. 1. X-ray crystal structure of ligand **8c** (ORTEP drawing).

1,4-bis-(diphenylphosphino)butane (10 mol%) and *i*-Pr₂NEt to afford **4** in 81–89% yields [43]. Hydrolysis of carboxylates **4** with aqueous KOH in methanol provided acids **5** in 93–99% yields after recrystallization from hot chloroform. Condensation of **5** with different enantiomerically pure 2-amino alcohols in the presence of 1-hydroxybenzotriazole (HOBt) and 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDCI) in DMF [44] gave amides **6** in 92–95% yields. The amides were subjected to oxazoline ring formation by treatment with triethylamine and methanesulfonyl chloride in the presence of a catalytic amount of 4-dimethylaminopyridine in dichloromethane [45,46]. Finally, target ligands **8** were obtained in 73–85% yields via a standard silane reduction using trichlorosilane and *N,N*-diisopropylethylamine in refluxing toluene. The molecular structure of **8c** was confirmed by single-crystal X-ray diffraction (Fig. 1) [47]. It is worthy to note that the ligands **8** are stable enough and can be purified by silica gel column chromatography. This characteristic is especially important for scale-up of the reactions.

3.2. Catalytic asymmetric addition of arylboronic acids to isatins

With ligands **8** prepared, we first examined their enantioselectivities for the palladium-catalyzed phenylboronic acid addition to benzyl-protected isatin along with some other ligands available in our laboratory. The results were summarized in Table 1. The reactions were performed by employing Pd(OAc)₂ as palladium source, 2-naphthol and LiOH·H₂O as additives. Interestingly, using (*S_aS*)-**8a** as the ligand, the reaction afforded the arylated product in 71% yield and 54% ee (Table 1, entry 1). Although higher catalytic activity was observed for (*S_aS*)-**8b** with an electron-rich cyclohexyl substituent, the enantioselectivity dropped sharply (Table 1, entry 2). Compared with the oxazoline ring bearing an *i*-Pr group, ligand (*S_aS*)-**8c** with a phenyl group provided better enantioselectivity of 66% ee (Table 1, entry 3 vs entry 1). The poor results for ligand (*S_aR*)-**8d** revealed the importance of chirality match of the oxazoline ring with the axially chiral biphenyl backbone (Table 1, entry 3 vs entry 4). Based on our initial screening results (Table 1, entries 1–4), we turned to examine the performances of ligands (*S_aS*)-**8e–8g** with various aryl groups linked to the P atoms, and ligand (*S_aS*)-**8g** with 3,5-di-*tert*-butyl phenyl groups furnished the highest activity and enantioselectivity for the Pd-catalyzed arylation of isatin (Table 1, entry 7). Ligand (*S_aS*)-**8e** and **8f** with less bulky 4-methylphenyl and 3,5-di-methylphenyl groups, exhibited slightly lower activities and enantioselectivities (Table 1, entries 5–7). Further fine-tuning the substituents R on the oxazoline ring ligands (*S_aS*)-**8h–8j** did not give better results (Table 1, entries 8–10), and (*S_aS*)-**8g** was the most effective ligand for the arylation reaction.

Scheme 1. Synthesis of ligand **8**.

Apparently, (*S_a,S*)-**8g** was also superior to its counterpart (*S_a,S*)-**9** with a binaphthyl skeleton for the isatin arylation, and higher yield of the arylation product along with a slightly better ee were obtained (Table 1, entry 7 vs 11). In addition to the

Table 1
Optimization experiments with different ligands.^a

Entry	Ligand	Yield (%) ^b	ee (%) ^c
1	8a	71	54
2	8b	81	3
3	8c	80	66
4	8d	28	47
5	8e	85	69
6	8f	83	70
7	8g	93	76
8	8h	79	55
9	8i	77	46
10	8j	73	44
11	9	82	73
12	10	38	40
13	11	12	43
14	12a	60	2
15	12b	21	11

^a Reaction conditions: 5 mol% Pd(OAc)₂, 10 mol% of ligand, LiOH·H₂O (1.0 equiv), 2-naphthol (0.15 equiv), room temperature, 48 h, THF.

^b Yield of isolated product.

^c The ee values were determined by HPLC with a Chiralcel OD-H column.

use of phosphine-oxazoline ligands, MeO-MOP (**10**) and some monophosphine ligands **11** [48] and **12a–12b** [49] developed recently by our group (Fig. 2) were also tested for this model reaction, but the results were disappointing (Table 1, entries 12–15). After surveying the experiment results, it is found that the reaction activity and enantioselectivity are greatly affected by the steric hindrance of the ligand. The biaryl framework of (*S_a,S*)-**8g**, including the bulky 3,5-di-tert-butylphenyl groups linked to the P atom and phenyl group on the oxazoline ring, provided an appropriate chiral environment so that the catalytic reaction acquired the highest yield and enantioselectivity. These findings highlight the importance of the combination of axially chiral biaryl phosphine skeleton and the adjustable modular oxazoline moiety for this reaction.

With (*S_a,S*)-**8g** as the ligand choice, further screening of the reaction conditions such as the palladium sources, bases and the needed amount, solvents, temperature and additives were undertaken. As listed in Table 2, conditional experiments showed that (*S_a,S*)-**8g**, THF, Pd(OAc)₂, 2-naphthol [40,50], and KOH were the best combination for this reaction. The effect of the ligand: metal mole ratios (1:1, 1.2:1, 2:1) were also tested, and it was found that the ligand: metal ratio of 1.2: 1 was enough for the reaction without loss of activity and enantioselectivity.

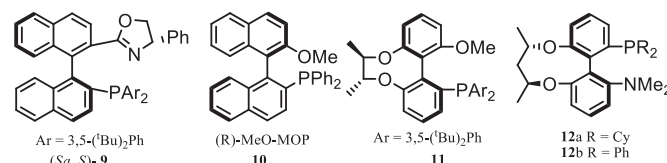
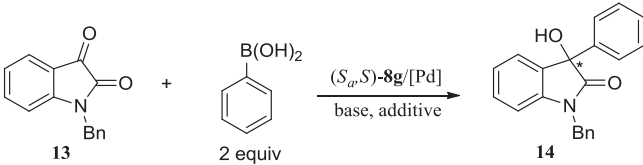
Fig. 2. Ligands **9–12**.

Table 2
Screening of Pd sources, bases, solvents and additives.^a

						
Entry	[Pd]	Base	Solvent	Additive	Yield (%) ^b	ee (%) ^c
1	Pd(OAc) ₂	LiOH·H ₂ O	THF	2-Naphthol	93	76
2	Pd ₂ (dba) ₃	LiOH·H ₂ O	THF	2-Naphthol	82	64
3	PdCl ₂	LiOH·H ₂ O	THF	2-Naphthol	32	51
4	Pd(TFA) ₂	LiOH·H ₂ O	THF	2-Naphthol	89	74
5	((η^3 -C ₃ H ₅) PdCl) ₂	LiOH·H ₂ O	THF	2-Naphthol	29	23
6	Pd(OAc) ₂	K ₂ CO ₃	THF	2-Naphthol	87	72
7	Pd(OAc) ₂	K ₃ PO ₄	THF	2-Naphthol	88	81
8	Pd(OAc)₂	KOH	THF	2-Naphthol	96	83
9	Pd(OAc) ₂	Cs ₂ CO ₃	THF	2-Naphthol	86	81
10	Pd(OAc) ₂	Et ₃ N	THF	2-Naphthol	62	83
11	Pd(OAc) ₂	KO ^t Bu	THF	2-Naphthol	85	78
12	Pd(OAc) ₂	DIPEA	THF	2-Naphthol	57	83
13	Pd(OAc) ₂	DBU	THF	2-Naphthol	0	ND
14	Pd(OAc) ₂	KOH	p-xylene	2-Naphthol	75	72
15	Pd(OAc) ₂	KOH	Toluene	2-Naphthol	81	75
16	Pd(OAc) ₂	KOH	Dioxane	2-Naphthol	80	74
17	Pd(OAc) ₂	KOH	CH ₂ Cl ₂	2-Naphthol	66	62
18	Pd(OAc) ₂	KOH	DMF	2-Naphthol	0	ND
19	Pd(OAc) ₂	KOH	DME	2-Naphthol	70	72
20	Pd(OAc) ₂	KOH	THF+ H ₂ O	2-Naphthol	<5	ND ^d
21	Pd(OAc) ₂	KOH	THF	1,2-Benzenediol	43	30
22	Pd(OAc) ₂	KOH	THF	Phenol	67	52
23	Pd(OAc) ₂	KOH	THF	(R)-BINOL	79	82
24	Pd(OAc) ₂	KOH	THF	(S)-BINOL	83	80
25	Pd(OAc) ₂	KOH	THF	4 Å MS	88	83
26	Pd(OAc) ₂	KOH	THF	1-Naphthol	66	51
27	Pd(OAc) ₂	KOH	THF	2,7-Dihydroxyl-naphthalene	82	79
28	Pd(OAc) ₂	KOH	THF	–	80	78 ^e
29	Pd(OAc) ₂	KOH	THF	2-Naphthol	80	73 ^f
30	Pd(OAc) ₂	KOH	THF	2-Naphthol	93	2 ^g
31	Pd(OAc) ₂	KOH	THF	2-Naphthol	83	73 ^h
32	Pd(OAc)₂	KOH	THF	2-Naphthol	95	83ⁱ
33	Pd(OAc) ₂	KOH	THF	2-Naphthol	85	76 ^j

^a Reaction conditions: 5 mol% Pd and 10 mol% of ligand, base (1.0 equiv), additive (0.15 equiv), room temperature, 48 h.

^b Yield of isolated product.

^c The ee values were determined by HPLC with a Chiralcel OD-H column.

^d Not determined, the ratio of THF and H₂O was 20:1.

^e 2-Naphthol was not added.

^f 0.5 equiv. base was used.

^g 2.0 equiv base was used.

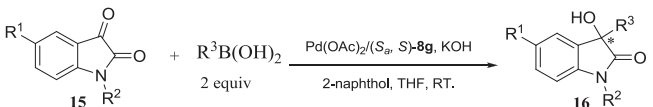
^h 45 °C, 24 h.

ⁱ 5 mol% Pd and 6 mol% of ligand.

^j 5 mol% Pd and 5 mol% of ligand.

With the optimal reaction conditions determined, the substrate scope was investigated. At first, we explored the reaction of methyl-, ethyl- or 1-trityl-protected isatin with phenylboronic acid, and only small amounts of the corresponding products were obtained. Further experiments showed that benzyl-protected isatins were appropriate substrates for the arylation reaction with various arylboronic acids. As summarized in Table 3, the reactions proceeded smoothly for arylboronic acids bearing electron-donating and electron-withdrawing groups. Substituents on the *para* and *meta* positions of the arylboronic acids were tolerated, and the corresponding adducts **16** were produced in high yields (up to 97%) and good enantioselectivities (up to 88% ee) (Table 3, entries 1–13). Facile arylation of benzyl-protected isatin with 2-naphthaleneboronic acid or 6-methoxy-2-naphthaleneboronic acid also occurred with excellent yields and good ee's (Table 3, entries 7 and 8). Arylboronic acids with *para*-substituents seemed

to give better enantioselectivities compared with those bearing *meta*-substituents (Table 3, entries 2, 3 and 11 vs entries 9, 10 and 12 respectively). Moreover, for substrate **15** bearing a more bulky 1-anthracen-9-ylmethyl, the arylation reaction with phenylboronic acid also happened, and a slight increase in enantioselectivity was observed albeit with a slight decrease in yield (Table 3, entry 14). Arylation of some halogenated or methyl-substituted isatins produced satisfactory results (Table 3, entries 15–17). It is noteworthy that the products can be easily enantioenriched to 98% enantiomeric purity by one recrystallization in ethyl acetate (Table 3, entry 2). Even for the product having a lower ee value such as 74% ee, similar enhancement of enantiopurity was achieved through the simple crystallization (Table 3, entry 7, 90% ee). In general, our catalytic system had obvious advantage over the Pd(OAc)₂ and biphenyl phosphinoimine combination in both catalytic activity and enantioselectivity [39]. In comparison with

Table 3Catalytic asymmetric arylation of isatins with arylboronic acids.^a


Entry	R ¹	R ²	R ³	Yield (%) ^b	ee (%) ^c
1	H	Bn	C ₆ H ₅	96	83
2	H	Bn	4-MeC ₆ H ₄	91 (80)	84 (98) ^d
3	H	Bn	4-MeOC ₆ H ₄	87	85
4	H	Bn	3,4-di-MeC ₆ H ₃	87	88
5	H	Bn	4-PhC ₆ H ₄	88	82
6	H	Bn	4- ^t BuC ₆ H ₄	80	76
7	H	Bn	2-naphthyl	97 (85)	74 (90) ^d
8	H	Bn	6-MeO-2-naphthyl	90	79
9	H	Bn	3-MeC ₆ H ₄	82	71
10	H	Bn	3-MeOC ₆ H ₄	91	73
11	H	Bn	4-ClC ₆ H ₄	85	80
12	H	Bn	3-ClC ₆ H ₄	90	78
13	H	Bn	4-CF ₃ OC ₆ H ₄	83	73
14	H	1-Anthracen-9-ylmethyl	C ₆ H ₅	72	86
15	Cl	1-Anthracen-9-ylmethyl	C ₆ H ₅	73	85
16	F	1-Anthracen-9-ylmethyl	C ₆ H ₅	70	84
17	CH ₃	1-Anthracen-9-ylmethyl	C ₆ H ₅	70	82

^a Reaction conditions: 5 mol% Pd and 6 mol% of ligand, KOH (1.0 equiv), 2-naphthol (0.15 equiv), 48 h.^b Yield of isolated product.^c Determined by HPLC with a Chiralcel OD-H or IA-3 or IC-3 column.^d After one recrystallization.

chiral cationic C₂-symmetric *N*-heterocyclic carbene(NHC) Pd²⁺ diaqua complex [40], the obtained results were comparable.

4. Conclusions

In conclusion, a series of new phosphine-oxazoline ligands were synthesized from commercially inexpensive BINOL in good yields. For the first time, a practical protocol for the enantioselective arylation of isatins with arylboronic acids was successfully realized by use of in situ formed catalysts generated from Pd(OAc)₂ and chiral phosphine-oxazolines. 3-Aryl-3-hydroxy-2-oxindoles were obtained in good yields (up to 97%) and enantioselectivities (up to 88% ee). Through a simple recrystallization, the enantiopurity of the products could be increased up to 98%. Further explorations of the chiral phosphine-oxazoline type ligands' applications in asymmetric catalysis are underway.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.apcata.2013.03.027>.

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